WEB DEVELOPMENT FRAMEWORKS

ITUE203

PRACTICAL 3

Responsive UI Design using CSS Grid, Flexbox, and Bootstrap

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Subject Code	ITUE203
Practical No.	Practical 3

Submission File

Objective

To design and demonstrate responsive web user interfaces using CSS Grid, CSS Flexbox, and Bootstrap. The practical should show how layouts adapt across different screen sizes such as desktop, tablet, and mobile devices.

Software / Tools Used

- Visual Studio Code
- HTML5
- CSS3
- Bootstrap 5
- Google Chrome / Microsoft Edge Developer Tools

1. CSS Grid

Aim

To create a responsive web layout using CSS Grid and demonstrate the use of rows, columns, gaps, fractional units, and media queries.

HTML Code

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Register - StudentHub</title>
    <link rel="stylesheet" href="../css/common.css" />
    <link rel="stylesheet" href="../css/grid.css" />
    <script src="../js/auth.js" defer></script>
  </head>
  <body data-auth-page="register">
    <header class="site-header">
      <div class="container">
        <h1><a href="../home.html">StudentHub</a></h1>
        <nav class="main-nav" aria-label="Main Navigation">
          <a href="../home.html">Home</a>
          <a href="features.html">Features</a>
          <a href="about.html">About</a>
          <a href="events.html">Events</a>
          <a href="contact-help.html">Contact/Help</a>
          <a href="login.html">Login</a>
          <a href="register.html" aria-current="page">Register</a>
        </nav>
      </div>
    </header>

    <main class="auth-container">
      <section class="page-title">
        <h2>Student Registration</h2>
        <p>Complete the form to demonstrate a responsive CSS Grid layout.</p>
      </section>

      <section class="auth-card">
        <p id="registrationMessage" class="form-message" role="alert" aria-
live="polite"></p>
        <form action="process.php" id="registrationForm" class="register-grid"
novalidate>
          <div class="form-group">
            <label for="first-name">First Name</label>
            <input id="first-name" name="first-name" type="text" required />
          </div>
```

```
          <div class="form-group">
            <label for="last-name">Last Name</label>
            <input id="last-name" name="last-name" type="text" required />
          </div>
          <div class="form-group">
            <label for="student-id">Student ID</label>
            <input id="student-id" name="student-id" type="text" required />
          </div>
          <div class="form-group">
            <label for="email">College Email</label>
            <input id="email" name="email" type="email" required />
          </div>
          <div class="form-group">
            <label for="phone">Phone Number</label>
            <input id="phone" name="phone" type="tel" required />
          </div>
          <div class="form-group">
            <label for="program">Program</label>
            <select id="program" name="program" required>
              <option value="">Select Program</option>
              <option>B.Tech Computer Engineering</option>
              <option>B.Tech Information Technology</option>
              <option>Bachelor of Computer Applications</option>
            </select>
          </div>
          <div class="form-group">
            <label for="semester">Semester</label>
            <select id="semester" name="semester" required>
              <option value="">Select Semester</option>
              <option>Semester 1</option><option>Semester 2</option>
              <option>Semester 3</option><option>Semester 4</option>
              <option>Semester 5</option><option>Semester 6</option>
              <option>Semester 7</option><option>Semester 8</option>
            </select>
          </div>
          <div class="form-group">
            <label for="password">Password</label>
            <input id="password" name="password" type="password" required />
          </div>
          <div class="form-group">
            <label for="confirm-password">Confirm Password</label>
            <input id="confirm-password" name="confirm-password" type="password"
required />
          </div>
          <div class="checkbox-group full-width">
            <input id="terms" name="terms" type="checkbox" required />
            <label for="terms">I agree to the terms and conditions.</label>
          </div>
```

```
          <div class="form-actions full-width">
            <button type="submit">Register</button>
            <button type="reset">Reset</button>
          </div>
        </form>
        <p>Already registered? <a href="login.html">Login here</a>.</p>
      </section>
    </main>

    <footer class="site-footer"><p>&copy; 2026 StudentHub. All Rights
Reserved.</p></footer>
  </body>
</html>
```

CSS Code

```
.dashboard-grid,
.summary-grid,
.register-grid,
.faculty-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 16px;
}

.dashboard-card,
.summary-card,
.faculty-card {
  padding: 20px;
  border: 1px solid #bdd8ef;
  border-radius: 8px;
  background-color: #f0f7ff;
}

.dashboard-card {
  color: #123d6a;
  font-weight: bold;
  text-align: center;
  text-decoration: none;
}

.dashboard-card:hover,
.dashboard-card:focus {
  color: #ffffff;
  background-color: #1769aa;
}

.summary-card {
  text-align: center;
```

```css
}

.summary-card h3,
.faculty-card h3 {
  margin-top: 0;
}

.summary-card p {
  margin-bottom: 0;
  font-size: 1.3rem;
  font-weight: bold;
}

.form-group label {
  display: block;
}

.checkbox-group label {
  font-weight: normal;
}

.full-width {
  grid-column: 1 / -1;
}

.form-actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
}

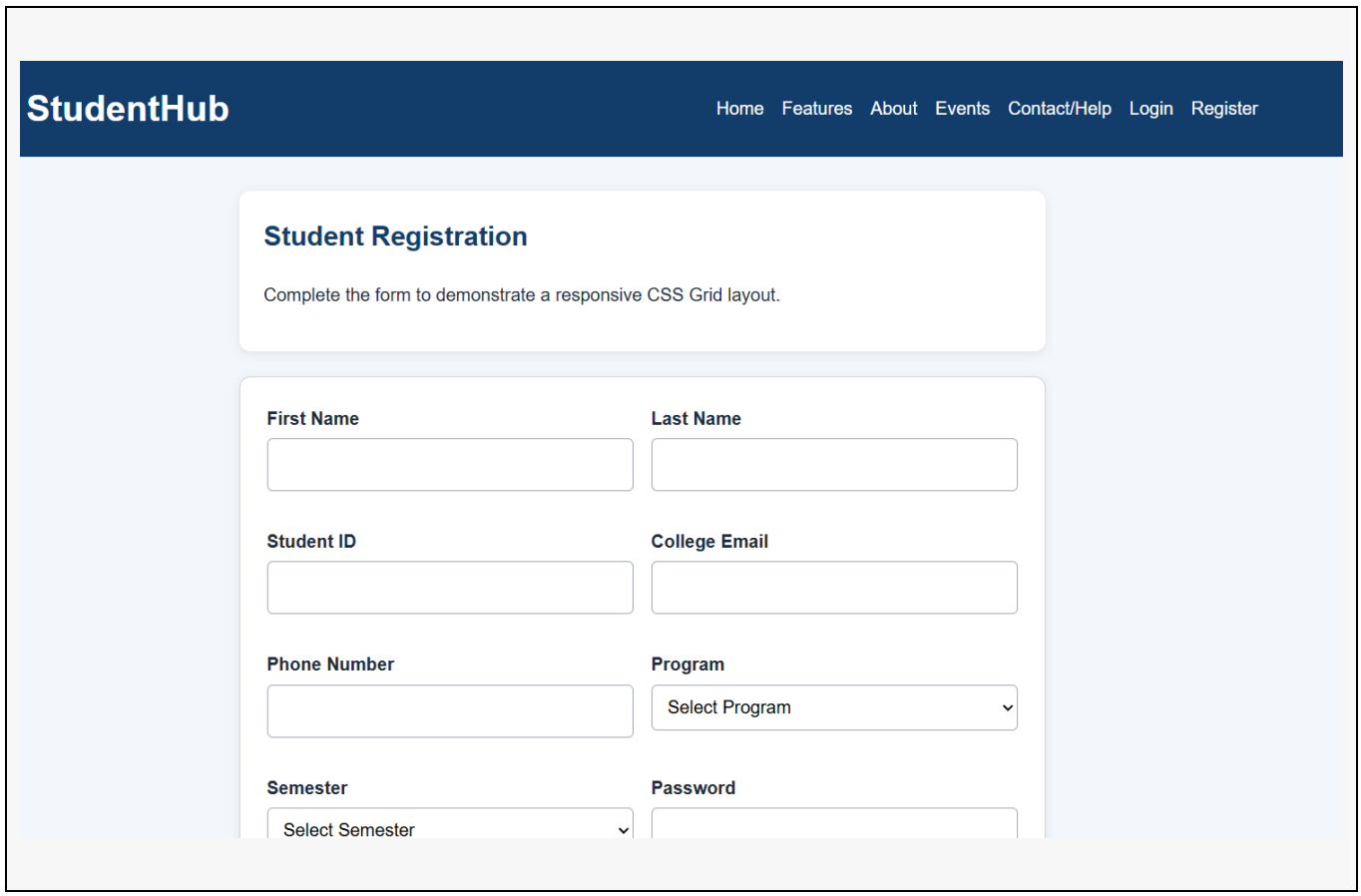
@media (min-width: 600px) {
  .dashboard-grid,
  .summary-grid,
  .register-grid,
  .faculty-grid {
    grid-template-columns: repeat(2, 1fr);
  }
}

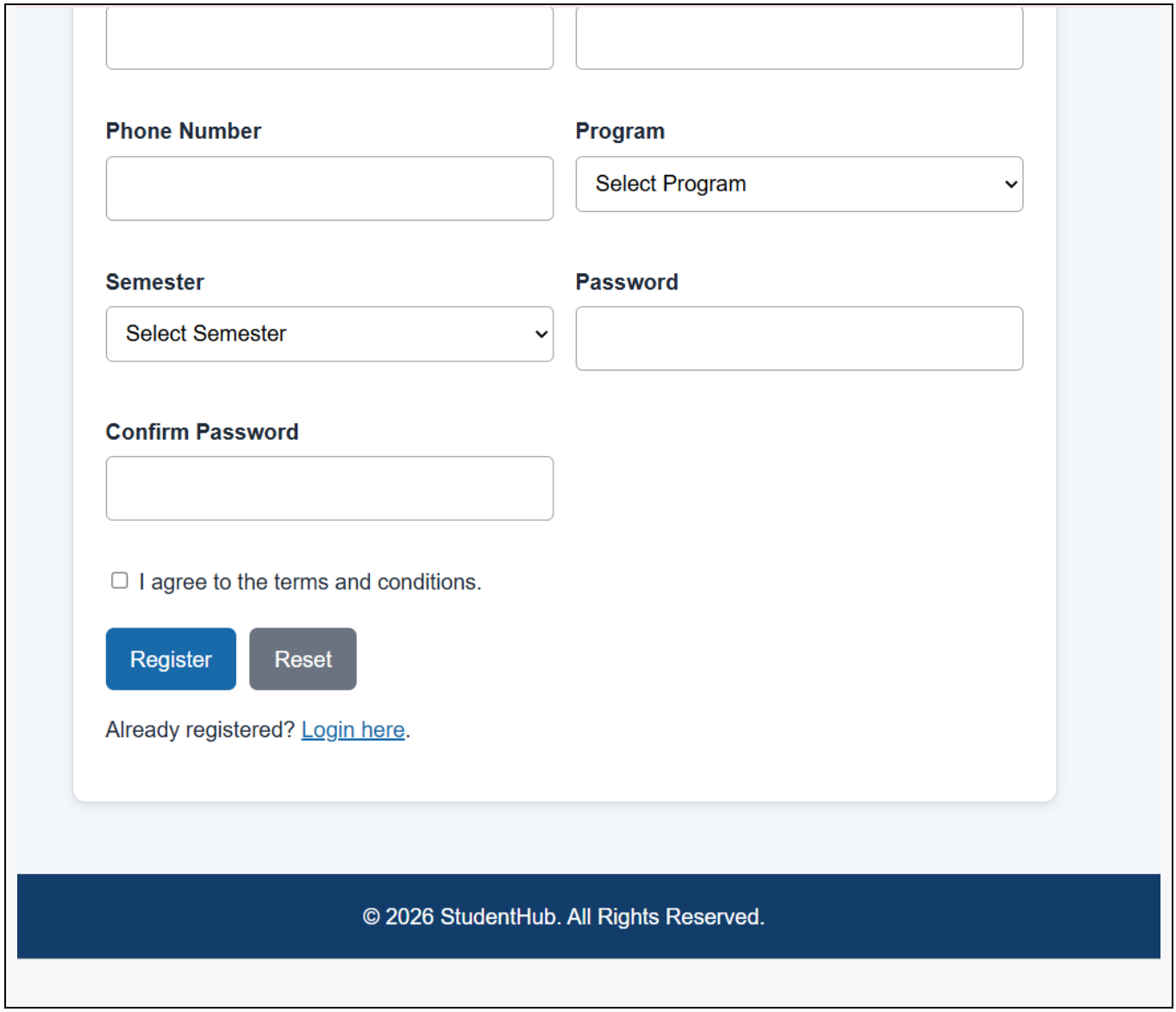
@media (min-width: 992px) {
  .dashboard-grid {
    grid-template-columns: repeat(3, 1fr);
  }

  .summary-grid {
    grid-template-columns: repeat(4, 1fr);
  }
```

```
  .faculty-grid {
    grid-template-columns: repeat(4, 1fr);
  }
}
```

Output Screenshot





Explanation

CSS Grid is used to create a structured and responsive two-dimensional layout by arranging content into rows and columns. The display: grid property enables the grid layout, while grid-template-columns defines the number and size of columns. The gap property is used to provide equal spacing between grid items. Fractional units such as 1fr divide the available space equally among columns, and the repeat() function makes the column definition shorter and easier to manage. Media queries are used to change the number of columns according to the screen size. On desktop screens, multiple cards can appear in one row, while on tablet screens the layout changes to fewer columns and on mobile screens the cards are displayed in a single column. This makes the webpage responsive and easy to use on different devices.

2. Flexbox

Aim

To create a responsive user interface using CSS Flexbox and demonstrate alignment, spacing, wrapping, and direction control.

HTML Code

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Home - StudentHub</title>
    <!-- <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB"
crossorigin="anonymous"> -->
    <link rel="stylesheet" href="css/common.css" />
    <link rel="stylesheet" href="css/flexbox.css" />
  </head>

  <body>
  <header class="site-header">
    <div class="container header-flex">
      <h1><a href="home.html">StudentHub</a></h1>

      <nav class="main-nav" aria-label="Main Navigation">
        <a href="pages/features.html">Features</a>
        <a href="pages/about.html">About</a>
        <a href="pages/contact-help.html">Contact/Help</a>
        <a href="pages/login.html">Login</a>
      </nav>
    </div>
  </header>

  <main class="container">
    <section class="hero-flex">
      <div>
        <h2>Welcome to StudentHub</h2>

        <p>
          StudentHub is an online academic portal where students can access
          courses, assignments, attendance, results, notices, timetable, and
          study resources.
        </p>
```

```
      </div>

      <a class="button" href="pages/register.html">Register Now</a>
    </section>

    <section>
      <h2>Quick Links</h2>

      <div class="quick-links">
        <a class="quick-card" href="pages/features.html">Features</a>
        <a class="quick-card" href="pages/about.html">About</a>
        <a class="quick-card" href="pages/contact-help.html">Contact / Help</a>
        <a class="quick-card" href="pages/login.html">Login</a>
        <a class="quick-card" href="pages/events.html">Events</a>
      </div>
    </section>
  </main>

  <footer class="site-footer">
    <p>&copy; 2026 StudentHub. All Rights Reserved.</p>
  </footer>
</body>
</html>
```

CSS Code

```
.header-flex,
.hero-flex,
.quick-links,
.about-flex,
.feature-flex {
  display: flex;
  flex-direction: column;
  gap: 16px;
}

.hero-flex {
  align-items: flex-start;
}

.quick-card,
.about-box,
.feature-box {
  flex: 1 1 200px;
  padding: 20px;
  border: 1px solid #bdd8ef;
  border-radius: 8px;
  background-color: #eaf3fc;
}
```

```css
.quick-card {
  color: #123d6a;
  font-weight: bold;
  text-align: center;
  text-decoration: none;
}

.quick-card:hover,
.quick-card:focus {
  color: #0f4f85;
  background-color: #d6e9f9;
}

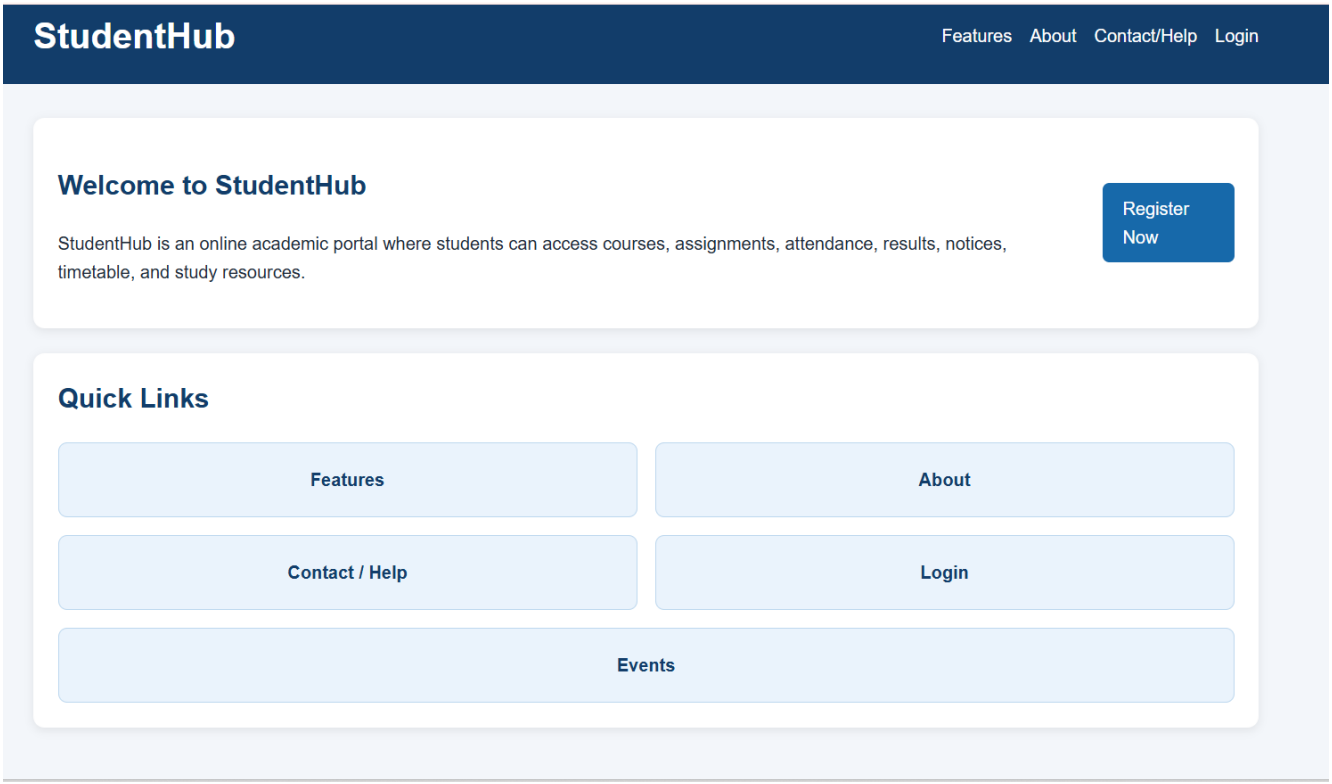
@media (min-width: 600px) {
  .quick-links,
  .about-flex,
  .feature-flex {
    flex-flow: row wrap;
  }

  .quick-card,
  .about-box,
  .feature-box {
    flex-basis: calc(50% - 16px);
  }
}

@media (min-width: 768px) {
  .header-flex,
  .hero-flex {
    flex-direction: row;
    align-items: center;
    justify-content: space-between;
  }
}

@media (min-width: 992px) {
  .about-box {
    flex-basis: calc(33.333% - 16px);
  }
}
```

Output Screenshot



Explanation

CSS Flexbox is used to arrange and align elements efficiently in a one-dimensional layout, either horizontally or vertically. The display: flex property converts a container into a flex container. The flex-direction property controls whether the items are arranged in a row or column. justify-content is used to control the alignment and spacing of items along the main axis, while align-items controls alignment along the cross axis. The gap property provides spacing between flex items, and flex-wrap allows items to move to the next line when sufficient space is not available. Media queries can also be used to change the flex direction on smaller screens. Therefore, Flexbox is useful for responsive navigation bars, content sections, buttons, and card layouts.

3. Bootstrap Responsive Design

Aim

To design a responsive web page using Bootstrap 5 components, utility classes, and the responsive grid system.

HTML Code

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Home - StudentHub</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB"
crossorigin="anonymous">
    <link rel="stylesheet" href="css/common.css" />
    <link rel="stylesheet" href="css/flexbox.css" />
  </head>

  <body class="bg-light text-dark">
    <div class="container py-4">
      <header class="mb-5">
        <h1 class="display-1 fw-bold text-primary mb-3">StudentHub</h1>

        <nav class="nav flex-wrap gap-3" aria-label="Main Navigation">
          <a class="nav-link text-primary fs-4 p-0"
href="pages/features.html">Features</a>
          <a class="nav-link text-primary fs-4 p-0"
href="pages/about.html">About</a>
          <a class="nav-link text-primary fs-4 p-0" href="pages/contact-
help.html">Contact/Help</a>
          <a class="nav-link text-primary fs-4 p-0"
href="pages/login.html">Login</a>
        </nav>
      </header>

      <main>
        <section class="mb-5">
          <h2 class="display-4 fw-bold text-dark mb-4">Welcome to StudentHub</h2>

          <p class="lead text-dark mb-4" style="max-width: 1200px;">
            StudentHub is an online academic portal where students can access
            courses, assignments, attendance, results, notices, timetable, and
```

```
            study resources.
          </p>

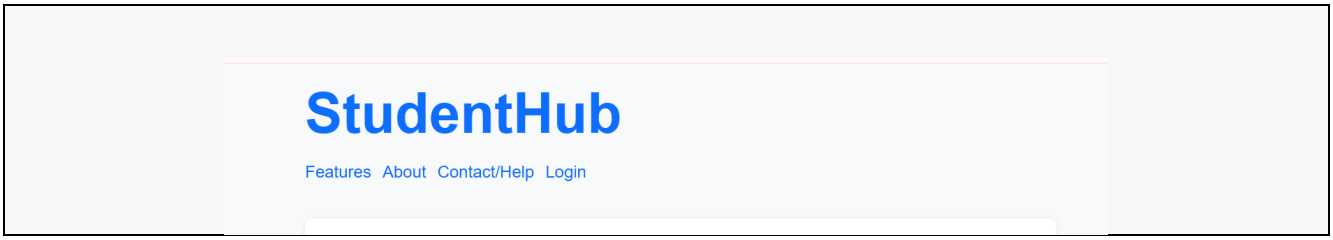
          <a class="btn btn-link text-primary fs-3 p-0 text-decoration-none"
href="pages/register.html">Register Now</a>
        </section>

        <section>
          <h2 class="display-5 fw-bold text-dark mb-4">Quick Links</h2>

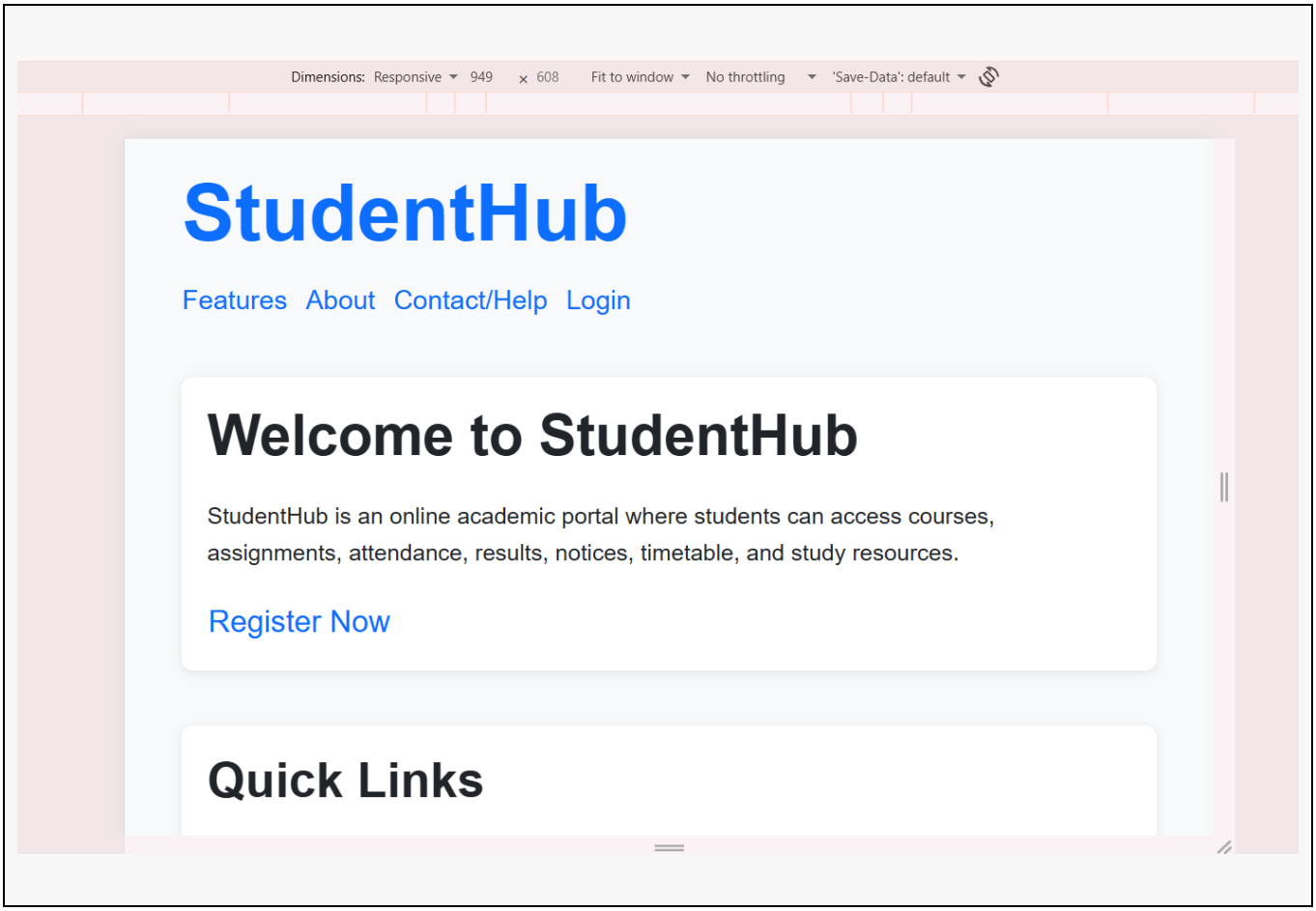
          <div class="d-flex flex-wrap gap-3">
            <a class="text-primary text-decoration-none fs-4"
href="pages/features.html">Features</a>
            <a class="text-primary text-decoration-none fs-4"
href="pages/about.html">About</a>
            <a class="text-primary text-decoration-none fs-4" href="pages/contact-
help.html">Contact / Help</a>
            <a class="text-primary text-decoration-none fs-4"
href="pages/login.html">Login</a>
            <a class="text-primary text-decoration-none fs-4"
href="pages/events.html">Events</a>
          </div>
        </section>
      </main>

      <footer class="mt-5 text-muted">
        <p class="mb-0 fs-5">&copy; 2026 StudentHub. All Rights Reserved.</p>
      </footer>
    </div>
  </body>
</html>
```

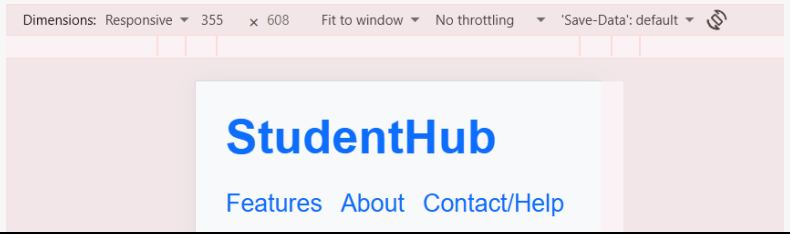
Desktop Output



Tablet Output



Mobile Output



4. Observation / Comparison

Technology	Main Purpose	Responsive Technique	Where Used
CSS Grid	Two-dimensional layout	Grid columns + media queries	Dashboard / card layout
Flexbox	One-dimensional alignment	Wrapping + direction changes	Navbar / content rows
Bootstrap	Responsive framework	Breakpoint-based grid classes	Complete responsive page

Conclusion

The practical demonstrates three responsive UI techniques. CSS Grid is suitable for structured two-dimensional layouts, Flexbox is effective for one-dimensional alignment and flexible component arrangement, and Bootstrap provides a ready-to-use responsive grid system and reusable UI components. The final web pages adapt correctly to desktop, tablet, and mobile screen sizes.